DG4COM Computational Intelligence

Aston University

Project: ACTIVEA (Physical Activity Planning Evolutionary Algorithm)

Scope: Core Task 1 (Data Setup) and Core Task 2 (Algorithm Setup and Model Formulation)

1. Executive Summary

This document covers Core Task 1 and Core Task 2.

- **Task 1 (Data Setup):** Builds the exercise-quality knowledge base of standardized physical activities, with intensity metrics calibrated for a 70 kg individual using ACSM guidelines. It also defines the stratified partitioning of the wearable telemetry dataset (`wearables_monitoring_data.csv`).

- **Task 2 (Algorithm Setup and Model Build):** Defines the evolutionary algorithm architecture: chromosome representation, selection, recombination and mutation operators, termination criteria, and two fitness formulations. These are the discrete lecture baseline (`fit.py.txt`) and an advanced continuous multi-attribute fitness function.

2. Task 1: Data Setup and Knowledge Base

2.1 Exercise Quality Knowledge Base

Activity	Calories/min (70 kg)	MET	Target RPE (1–10)	Target HR (% MHR)	Fitness domains
Jogging	7.35	6.0	5.0	70 – 77%	CV, ME, FM
Bodyweight squats	4.90	4.0	6.0	64 – 70%	ST, ME, BA, FM
Cycling (moderate)	6.80	5.5	4.0	65 – 75%	CV, ME
Plank	3.50	3.0	7.0	60 – 68%	ST, ME, BA
Yoga	3.00	2.5	3.0	55 – 65%	FL, BA, FM
Rope skipping	11.00	9.0	8.0	78 – 88%	CV, AG, ME, FM

Running	10.50	8.5	7.0	77 – 85%	CV, ME, FM
Brisk walking	4.10	3.5	3.0	55 – 65%	CV, FM
Pushups and calisthenics	5.60	4.5	6.0	65 – 75%	ST, ME, FM
Agility drills and burpees	9.80	8.0	8.0	80 – 90%	CV, AG, ST, FM

2.2 Metric Definitions and References

Calories burned per minute (kcal/min). For a standard 70 kg adult, energy expenditure is derived from METs using the ACSM (2018) metabolic equation:

```
Calories/min = (MET × 3.5 × Weight_kg) / 200
```

For 70 kg this reduces to `Calories/min = MET × 1.225`. For example, jogging (MET = 6.0) gives 6.0 × 1.225 = 7.35 kcal/min.

Metabolic Equivalent of Task (MET). The energy cost of an activity as a multiple of resting metabolic rate (1 MET ≈ 3.5 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$). Values are cross-referenced with the 2011 Compendium of Physical Activities (Ainsworth et al., 2011; Jetté et al., 1990).

Target Rate of Perceived Exertion (RPE). Measured on the Borg CR10 scale (Borg, 1998), from 1 (very light) to 10 (maximal). The target RPE is the planned effort of ceiling to safely sustain the workout.

Target heart rate (% MHR). Percentage of maximum heart rate (MHR ≈ 220 − age, or the Karvonen heart-rate-reserve method). It maps to ACSM aerobic training zones:

- Moderate aerobic endurance: 64–76% MHR
- Vigorous cardiorespiratory conditioning: 77–93% MHR

Seven fitness domains.

Code	Domain	Description	Examples
CV	Cardiovascular fitness	Sustained aerobic oxygen delivery	Jogging, cycling, running
ST	Strength development	High-force muscular resistance	Squats, pushups
ME	Muscular endurance	Sustained submaximal contractions	Plank, squats
FL	Flexibility / mobility	Full usable range of joint motion	Yoga

BA	Balance and stability	Static and dynamic core stability	Plank, yoga
AG	Agility	Rapid multidirectional acceleration and deceleration	Rope skipping, agility drills
FM	Functional movement	Natural compound movement mechanics	Squats, walking, burpees

2.3 Wearable Telemetry Dataset Partitioning

The empirical dataset (`wearables_monitoring_data.csv`) holds 10,000 physiological observations from wearable sensors across six activity regimes: Lying, Sitting, Self Pace walk, running 3 METs, running 5 METs and Running 7 METs.

To validate models without data contamination:

- **Stratified split:** records are stratified proportionally across the six activity classes.
- **Training partition (80%):** N_train = 7,998 records, used to calibrate baseline user profiles and Karvonen relative intensities.
- **Testing partition (20%):** N_test = 2,002 records, held out for out-of-sample evaluation of evolved workout recommendations.

3. Task 2: Algorithm Setup and Model Formulation

3.1 Algorithm Selection: Genetic Algorithm

A Genetic Algorithm (GA) was chosen because exercise session planning is a combinatorial subset-and-parameter allocation problem: select m activities and assign each a continuous duration. GAs balance global exploration of activity combinations with local exploitation of duration allocations.

3.2 Chromosome Representation

A workout session is a variable-length sequence of (activity, duration) tuples:

```
S = [ (a1, d1), (a2, d2), ..., (am, dm)]
```

- **Activity allele ($a_i \in A$):** a unique activity from the knowledge base.
- **Duration allele ($d_i \in [5.0, 45.0]$ min):** execution time in minutes.
- **Chromosome length ($m \in [2, 5]$):** at least 2 to prevent single-activity monotony, at most 5 to avoid over-fragmentation.

3.3 Genetic Operators

Selection

- Tournament selection (k = 3): draw 3 individuals at random and keep the fittest. This controls selection pressure and limits premature convergence.
- Elitism (E = 2): the 2 best solutions of generation t pass unchanged to generation t+1, so best fitness never decreases.

Recombination (crossover)

Uniform recombination with deduplication. For parents P1 and P2, activities are merged and partitioned. If the same activity appears more than once, durations are summed:

`(a_i, d_i,1) + (a_i, d_i,2) → (a_i, d_i,1 + d_i,2)`

Mutation (p_m = 0.15)

- Duration perturbation: add $\Delta d \in \{-10, -5, -2, +2, +5, +10\}$ min, clipped to [5.0, 45.0].
- Activity substitution: replace an activity with one not currently in the session.
- Structural addition/deletion: grow or shrink the session within [2, 5] items.

3.4 Fitness Function Formulation

Model A: Lecture Baseline (fit.py.txt / Code 2)

The lecture baseline uses a discrete step score: +1 when a target is met, −1 otherwise.

```
calc_fitness(actv_session) returns fit_val := 0:
     # 1. Caloric constraint (healthy range 300 to 600 calories)
     if (calories >= 300 AND calories <= 600): fit_val += 1
     else: fit_val -= 1

    # 2. MET intensity constraint
     if (MET >= 4.0): fit_val += 1
     else: fit_val -= 1

    # 3. Target RPE constraint
     if (RPE >= 4.0 AND RPE <= 7.0): fit_val += 1
     else: fit_val -= 1

    # 4. Target heart rate constraint
```

```
if (HR >= 64.0 AND HR <= 80.0): fit_val += 1
else: fit_val -= 1
```

Analysis. This scoring is simple but yields a plateau landscape. Fitness is restricted to the integers {−4, −2, 0, +2, +4}, and chromosomes near a boundary receive no gradient information.

Model B: Continuous Multi-Attribute Fitness

To give the search a smooth gradient, ACTIVEA uses a continuous, 100-point normalized objective:

```
F(S) = max (0, w_cal·f_cal + w_met·f_met + w_rpe·f_rpe + w_hr·f_hr +
w_dom·f_dom − P_dur(S) − P_struct(S))
```

Each component score f_i lies in [0, 1].

Component	Weight	Formula
Caloric score	30	f_cal = max (0, 1 − abs (C_total − C_target) / (C_max − C_min))
MET intensity	15	f_met = max (0, 1 − abs (MET_avg − MET_target) / ΔMET)
RPE effort	15	f_rpe = max (0, 1 − abs (RPE_avg − RPE_target) / ΔRPE)
Heart rate alignment	15	f_hr = max (0, 1 − abs (HR_avg − HR_target) / ΔHR)
Domain diversity	25	f_dom = (domains covered) / 7

Penalties. P_dur subtracts 1.5 points per minute that total session duration falls outside [20, 75] minutes. P_struct subtracts 10 points for mono-activity routines.

3.5 Termination Criteria

Following `Code 1 (DG4COMwk1psd_ACTIVEA.py.txt)`:

```
Terminate: = (Fitness(S_best) ≥ τ) OR (Generation ≥ G_max)
```

- **Quality threshold (τ = 95.0):** stops evolution early once a near-perfect workout is found.
- **Maximum generations (G_max = 60):** bounds computation, with runs completing in under 0.5 seconds.

4. Codebase Architecture and File Mapping

Coursework component	Implementation
Activity and session models	`activea/models.py`
Task 1: activity table and dataset loader	`activea/data_loader.py`
Task 2: fitness evaluators (baseline and continuous)	`activea/fitness.py`
Task 2: selection, crossover and mutation	`activea/operators.py`
Tasks 2/3: evolutionary execution loop	`activea/core_ga.py`
Verification and unit tests	`tests/test_all.py`

5. Review Notes (check before submitting)

Calorie values vs. the stated formula. Several table entries do not match MET × 1.225:

Activity	MET	Table value	Formula value
Cycling (moderate)	5.5	6.80	6.74
Plank	3.0	3.50	3.68
Yoga	2.5	3.00	3.06
Running	8.5	10.50	10.41
Brisk walking	3.5	4.10	4.29
Pushups and calisthenics	4.5	5.60	5.51

Jogging, squats, rope skipping and burpees already match. Either correct the table or state that the remaining values come from a different source.

Split sizes. 7,998 / 2,002 is not exactly 80/20 of 10,000 (8,000 / 2,000). If this comes from per-class rounding in the stratified split, add a sentence saying so.

Elitism wording. With E = 2 best fitness is non-decreasing, not strictly monotonic, because it can stay flat across generations.

Fitness targets. Model B uses C_target, MET_target, RPE_target, HR_target and the Δ ranges without giving values. Listing to them in a small table would make the formulation reproducible.

Dataset Link: wearables_monitoring_data.csv